Year 13 Further Maths - Centres of mass

Similar questions - Composite bodies and suspension

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 $ABCD$ is a uniform lamina in the shape of a kite with $AC = 0.8$ m, BD meeting AC perpendicularly at its midpoint M, $BM = 0.3$ m and $DM = 0.9$ m.

(a) Explain why the centre of mass G lies on BD. **[1]**

(b) Show that the distance of G from B is 0.5 m. **[4]**

(c) The lamina is freely suspended from A. Find the acute angle that AC makes with the vertical. **[4]**

2

(a) A circular sector of radius r has angle 2θ. The segment cut off by the chord has its centre of mass at distance $\dfrac{2r\sin^3\theta}{3(\theta - \sin\theta\cos\theta)}$ from the centre. For $\theta = \frac{\pi}{3}$ and $r = 4$, find the distance of the centre of mass of the segment from the centre and its area. **[3]**

(b) A uniform circular lamina of radius 4 cm has centre O at the origin. A segment, formed by the chord at distance 2 cm from O and the arc beyond it, is removed with the axis of symmetry along the positive y-axis. Find the coordinates of the centre of mass of the remaining shape. **[5]**

(c) The remaining shape is hung freely from the point $(4, 0)$ on its rim. Find the angle between the line of symmetry and the vertical. **[4]**

3 A shape S is formed by attaching a particle of mass $2m$ to the vertex of a uniform solid cone of mass $8m$, height h and base radius 1.1 m. Two vertical strings support S in equilibrium with its axis horizontal: an inextensible string of length 1.6 m attached to A, the highest point of the rim of the base, and an elastic string of modulus $8mg$ attached to the vertex. Both strings are attached to a horizontal ceiling.

(a) Explain why the centre of mass of S lies on the axis. **[1]**

(b) Show that the tension in the elastic string is $4mg$. **[4]**

(c) Determine the natural length of the elastic string. **[4]**

Question 1

(a) The kite is symmetrical about BD, so G lies on BD

(b) Triangle ABC: area 0.12, centroid 0.1 from M towards B. Triangle ACD: area 0.36, centroid 0.3 from M towards D. Taking $B \to D$ positive from M: $\bar{y} = \dfrac{0.12(-0.1)+0.36(0.3)}{0.48} = 0.2$ from M, so $BG = 0.3+0.2 = 0.5$ m

(c) G is 0.2 m from M on MD; when suspended from A, AG is vertical. $\tan\widehat{GAM} = \dfrac{0.2}{0.4}$, so AC makes $\tan^{-1} 0.5 = 26.6°$ with the vertical

Question 2

(a) $\theta - \sin\theta\cos\theta = \frac{\pi}{3} - \frac{\sqrt{3}}{4} = 0.6142$; area $= r^2(\theta - \sin\theta\cos\theta) = 9.83$ cm^2. Distance $= \dfrac{2(4)\left(\frac{\sqrt{3}}{2}\right)^3}{3(0.6142)} = 2.82$ cm

(b) Disc area $16\pi = 50.27$, centre 0; segment area 9.83 at $y = 2.82$. $\bar{y} = \dfrac{-9.83(2.82)}{50.27-9.83} = -0.685$. Centre of mass $(0, -0.685)$

(c) G lies on the axis so PG is vertical, where $P = (4,0)$; the angle between the axis (the y-axis) and PG is $\tan^{-1}\dfrac{4}{0.685} = 80.3°$

Question 3

(a) The cone and the particle are both symmetric about the axis, so G lies on it

(b) Measure from the base. Cone: $\frac{h}{4}$, mass $8m$; particle h, mass $2m$. $\bar{x} = \dfrac{8m\frac{h}{4}+2mh}{10m} = \dfrac{2}{5}h$. Moments about A: $T_2 h = 10mg \times 0.4h$, $T_2 = 4mg$

(c) Extension $e = \dfrac{T_2 l_0}{\lambda} = \dfrac{4mg\,l_0}{8mg} = \frac{1}{2}l_0$. Total vertical distance from ceiling to the vertex is $1.6+1.1 = 2.7$ m $= 1.5l_0$, so $l_0 = 1.8$ m